

Real analysis Qualifying exam, August 2020

Make sure that you have signed the Honor Pledge on Collab.

In order to receive the full credit for a problem, a detailed argument (rather than a sketch of the proof) is needed. Whenever applying one of the standard theorems, please indicate that clearly.

Problem 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, almost everywhere differentiable, and such that $f'(x) = 1$ almost everywhere. (Both "almost everywhere" properties are assumed with respect to the Lebesgue measure on $\mathbb{R}$.) Does this imply that $f(2) - f(1) = 1$?

If yes, prove this. If no, give a counterexample.

Problem 2

Is every open set in $\mathbb{R}^2$ a countable union of closed sets?

If yes, prove this. If no, give a counterexample.

Problem 3

Let $\mathcal{H}$ be a separable complex Hilbert space with basis (complete orthonormal system) $f_1, f_2, f_3, \dots$. Define a linear operator P in $\mathcal{H}$ by setting

$$P(f_n) = f_{n+1}, \quad n = 1, 2, \dots$$

- (a) Find the adjoint P^* to P .
- (b) Find the operators PP^* and P^*P .

Problem 4

Let $(X, \mathcal{F}, \mu)$ be a measure space with $\mu(X) = 1$. Let $f_n : X \rightarrow \mathbb{R}$ be measurable functions such that for all $t \in \mathbb{R}$,

$$\lim_{n \rightarrow +\infty} \mu(x : f_n(x) \leq t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Show that $f_n \rightarrow 0$ in measure.

Problem 5

Show that the operator

$$(Tf)(x) := \int_0^\infty \frac{f(y)}{x+y} dy$$

is bounded in the space $L^p(\mathbb{R}_{\geq 0})$ for all $1 < p < +\infty$.